



1 billion metric tons of food wasted

In 2019, United Nations report shows that 1 billion metric tons of food was wasted. <https://digit.in/apr21-45>



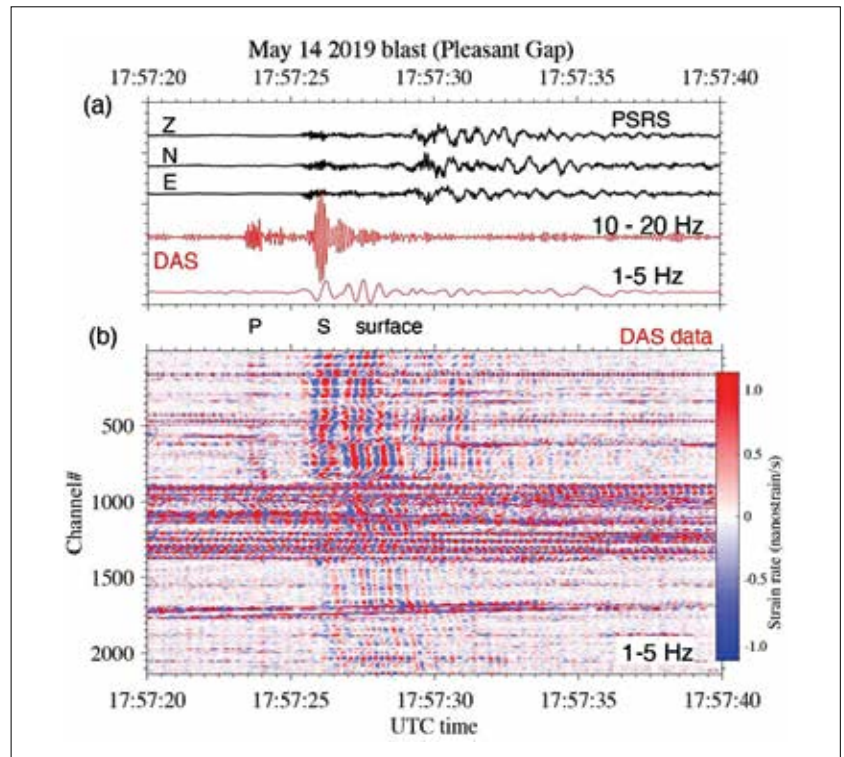
Green Burials

Even though it is slowly gaining popularity, most people are unaware of Green Burials as deathcare. <http://digit.in/apr21-46>

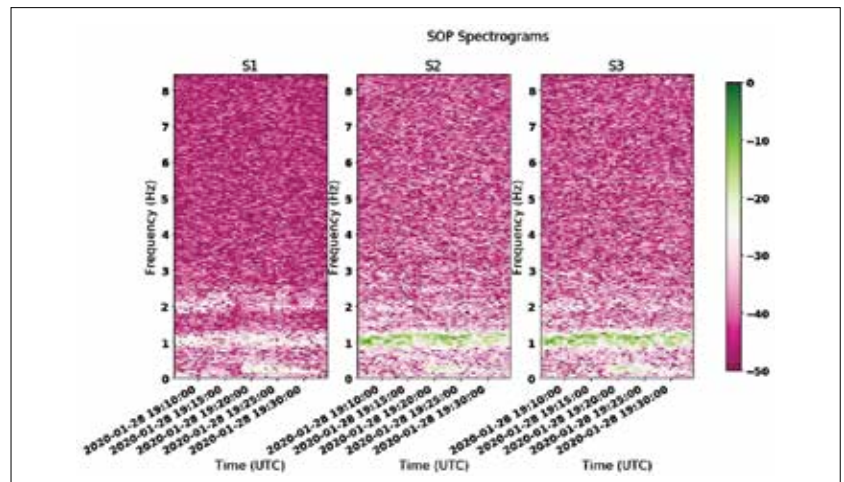
registered a magnitude 7.7 earthquake off the coast of Jamaica.

Researchers from Penn State tapped into dark fibre to test their implementation of distributed acoustic sensing, one of the many such experiments in the area. The cable length used spanned the city, and was just under 5 kilometers long, a stretch that was converted to 2,300 sensors. Terabytes of high resolution data was generated and the researchers could pick apart everything from crowds walking, to thunderstorms, to industrial activity. The data collected by the geophysicists has a number of promising future applications, including tracking traffic movement in a city without having to use data from phones, monitoring industrial noise pollution, and detecting smaller scale underground movements that lead to sinkholes and flooding. The technology is sensitive enough to detect mining blasts, people walking, gunshots, and even a particular song playing at a concert. One of the promising future applications of the technology is to deploy it in harsh environments, such as for monitoring the stability of permafrost in the Arctic. The experiment then began to detect a series of earthquakes over the next few months. Caltech researchers verified the detections, and pointed out that the data could not only be used to detect earthquakes, but also changes in the pressure of the ocean, which in turn can be used to detect tsunamis.

Zhongwen Zhan, assistant professor of geophysics at Caltech who confirmed the detections by the engineers at Google, began to refine the methods being used, and along with his colleagues, developed a method of using the data from the lit cables for geosensing applications. The approach allows the use of undersea cables actively being used as part of the communications infrastructure, without needing too much additional cost or equipment. Zhan says, "This new technique can really convert the majority of submarine cables into geophysical sensors that are thousands of kilometers long to detect earthquakes and possibly



A mining blast captured by the FORSEE network



The Google team registering the earthquake in Jamaica

tsunamis in the future. We believe this is the first solution for monitoring seismicity on the ocean floor that could feasibly be implemented around the world. It could complement the existing network of ground-based seismometers and tsunami-monitoring buoys to make the detection of submarine earthquakes and tsunamis much faster in many cases." The researchers detected many earthquakes, but no tsunamis. They did however register

ocean swells, which demonstrates that in principle, the technology should be able to detect tsunamis as well. The researchers are now looking to develop machine learning algorithms that can interpret the characteristics of the light in undersea cables, to provide realtime geosensing data. These networks can potentially be used to complement existing networks of sensors to monitor seismic activity, both on the land and under the sea. 